

Wazuh Agent Deployment Procedure

Purpose

Wazuh agents extend endpoint visibility across the Virtual SOC environment. They provide: - File integrity monitoring (FIM) - Authentication logging - Process and command monitoring - Vulnerability detection - Malware detection logic - System inventory and configuration changes

Agents run on: - Victim hosts - Metasploitable 3 (optional with manual install) - Kali attacker (optional visibility) - Utility VMs (RDP client, internal Ubuntu hosts)

1. Prerequisites

- Wazuh Manager installed and reachable
 - Agent host with network access to Manager on ports **1514/UDP** and **1515/TCP**
 - Root or administrator privileges on the endpoint
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2. Deployment Workflow

The process is handled via the Wazuh Dashboard, which generates host-specific install commands.

Step 1 — Open Dashboard

Navigate to:

`https://<wazuh-ip>:443`

Login with admin credentials.

Step 2 — Navigate to Agent Deployment

Dashboard:

Agents → Deploy New Agent

Step 3 — Configure Agent

You must enter: - **Server address** (Wazuh Manager IP) - **Agent name** (hostname or role) - **Operating system**

Dashboard generates installation commands automatically.

3. Installation Commands

Linux

Example generated script:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent_<version>.deb
sudo dpkg -i wazuh-agent_<version>.deb
sudo /var/ossec/bin/agent-auth -m <wazuh-manager-ip> -A <agent-name>
sudo systemctl enable --now wazuh-agent
```

Windows

The dashboard provides an MSI installer and agent-auth details.

macOS

The dashboard provides a pkg installer.

4. Registration Validation

Ensure the agent has registered correctly.

On endpoint:

```
sudo systemctl status wazuh-agent
```

Expected active state.

On Manager (Dashboard):

Agents → Agent List

Status transitions: - **Never connected** → pre-auth - **Pending** → key exchange in progress - **Active** → agent fully operational

5. Telemetry Forwarded by Agents

Security Logs

- auth.log / system.log
- Windows Event Logs
- Login failures, sudo abuse, brute-force indicators

File Integrity Monitoring (FIM)

Real-time monitoring of directories such as: - /etc - /var/log - /home

Vulnerability Detection

Automatic CVE scanning using OS build and package lists.

Process Monitoring

- Unexpected root processes
- Execution of dangerous binaries

System Inventory

- Installed packages
 - Kernel version
 - Network interfaces
-

6. Testing Agent Functionality

Test 1 — Service Stop

```
sudo systemctl stop ssh
```

Wazuh should alert on service interruption.

Test 2 — Authentication Failure

```
sudo su - invaliduser
```

Dashboard receives failed authentication attempts.

Test 3 — File Integrity

Modify /etc/hosts:

```
sudo nano /etc/hosts
```

Wazuh triggers file integrity alert.

Test 4 — Vulnerability Scan

Automatically triggered at agent startup. Scan results appear under:

```
Security events → Vulnerabilities
```

7. Agent Management

Restart Agent

```
sudo systemctl restart wazuh-agent
```

Remove Agent from Manager

Dashboard:

```
Agents → Select agent → Remove
```

Re-register Agent

```
sudo /var/ossec/bin/agent-auth -m <manager-ip> -A <agent-name>
```

8. Integration in the VSOC

Agents are installed on all instructional systems, providing: - Full EDR telemetry - Visibility into attacker activity on compromised machines - System changes on internal hosts - Combined analysis with Suricata network alerts

This dual host/network data model enables accurate attack-chain correlation inside the Virtual SOC.